Python Activity-Based Learning

Programs 1-10: Sample Inputs and Outputs

This document contains one sample run for each of the ten Python programs supplied in the conversation. Inputs and calculated results follow the example policies used in those programs.

These are formatted console transcripts, not screenshots of terminal executions. Prompts and spacing are condensed in places for readability; the numerical results are calculated from the corresponding program rules.

All tariffs, exchange rates, eligibility thresholds, and concession policies are illustrative classroom assumptions. They are not official financial, hospital, railway, or telecommunications policies.

1. University Scholarship Allocation
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3. Hospital Billing and Patient Classification
4. Employee Payroll and Deduction Analysis
5. Multi-Plan Telecommunications Cost Optimization
6. Railway Fare and Reservation Calculator
7. Retail Billing with Multiple Discount Rules
8. Manufacturing Cost and Profitability Analysis
9. Multi-Currency Travel Budget Analysis
10. Campus Event Financial Planning

1. University Scholarship Allocation

Two scholarships; includes a ranking tie and a subject-wise rejection.

```
Number of applicants: 4
Number of scholarships available: 2

Registration ID: S101
Name: Asha
Subject marks: [90, 85, 88]
Attendance percentage: 92
Annual family income: 180000
Eligible

Registration ID: S102
Name: Ravi
Subject marks: [85, 90, 88]
Attendance percentage: 95
Annual family income: 250000
Eligible

Registration ID: S103
Name: Meena
Subject marks: [78, 80, 82]
Attendance percentage: 85
Annual family income: 200000
Eligible

Registration ID: S104
Name: Kiran
Subject marks: [90, 55, 95]
Attendance percentage: 90
Annual family income: 180000
Not eligible:
- Subject 2 below 60

SCHOLARSHIP RANKING
Rank | ID | Name | Aggregate | Status
1 | S102 | Ravi | 87.67 | Selected
2 | S101 | Asha | 87.67 | Selected
3 | S103 | Meena | 80.00 | Waiting list
```

2. Progressive Electricity Billing

Three months demonstrate a rebate, progressive slabs, and a surcharge.

```
Number of months: 3

Month name: January
Units consumed: 80
Energy charge: Rs. 120.00
Fixed charge: Rs. 100.00
Surcharge: Rs. 0.00
Rebate: Rs. 6.00
Final bill: Rs. 214.00

Month name: February
Units consumed: 180
Energy charge: Rs. 350.00
Fixed charge: Rs. 100.00
Surcharge: Rs. 0.00
Rebate: Rs. 0.00
Final bill: Rs. 450.00

Month name: March
Units consumed: 350
Energy charge: Rs. 1000.00
Fixed charge: Rs. 100.00
Surcharge: Rs. 100.00
Rebate: Rs. 0.00
Final bill: Rs. 1200.00

MONTHLY COMPARISON
January | Units: 80.0 | Bill: Rs. 214.00
February | Units: 180.0 | Bill: Rs. 450.00
Change from previous month: 110.28 %
Alert: unusual bill increase
March | Units: 350.0 | Bill: Rs. 1200.00
Change from previous month: 166.67 %
Alert: unusual bill increase

Total bills: Rs. 1864.00
Average bill: Rs. 621.33
Highest bill month: March
```

3. Hospital Billing and Patient Classification

The larger 15% low-income concession applies instead of stacking concessions.

```
Number of patients: 1

Patient 1
Patient name: Lakshmi
Age: 65
Annual family income: 180000
Consultation charge: 500
Diagnostic charge: 2000
Treatment charge: 5000

Gross bill: Rs. 7500.00
Concession: Rs. 825.00
Final bill: Rs. 6675.00
Amount paid: 5000

PATIENT BILL
Name: Lakshmi
Payment classification: Partially paid
Paid: Rs. 5000.00
Outstanding: Rs. 1675.00
```

4. Employee Payroll and Deduction Analysis

Monthly payroll using the classroom deduction policy, not statutory tax rates.

```
Number of employees: 1
Employee ID: E101
Employee name: Arun
Basic salary: 40000
Overtime hours: 10
Incentive: 3000

========================================
SALARY STATEMENT
Employee: E101 - Arun
----------------------------------------
Basic salary                   40000.00
HRA                             8000.00
DA                              4000.00
Overtime pay                    2000.00
Incentive                       3000.00
Gross salary                   57000.00
Provident fund                  4800.00
Progressive deduction           1950.00
Total deductions                6750.00
Net salary                     50250.00
========================================
```

5. Multi-Plan Telecommunications Cost Optimization

All plans cover the requested validity; cost determines eligibility and selection.

```
Total required call minutes: 500
Total required data in GB: 8
Required validity in days: 28
Maximum budget: 800

PLAN COMPARISON

Plan: Basic
Validity: 28 days
Excess charges: Rs. 330.00
Total cost: Rs. 529.00
Eligible

Plan: Standard
Validity: 28 days
Excess charges: Rs. 0.00
Total cost: Rs. 399.00
Eligible

Plan: Premium
Validity: 56 days
Excess charges: Rs. 0.00
Total cost: Rs. 699.00
Eligible

Plan: LongValid
Validity: 84 days
Excess charges: Rs. 0.00
Total cost: Rs. 999.00
Not eligible: exceeds budget

Most economical eligible plan: Standard
Cost: Rs. 399.00
```

6. Railway Fare and Reservation Calculator

Illustrative fares; one senior ticket is cancelled and one child ticket is retained.

```
Journey distance in km: 600
Class (sleeper/ac/first): ac
Number of passengers: 2

Passenger 1
Name: Ramesh
Age: 65
Travel fare: Rs. 576.00
Reservation fee: Rs. 40.00
Ticket total: Rs. 616.00
Cancel this ticket? (yes/no): yes
Hours remaining before departure: 60
Cancellation penalty: Rs. 57.60
Refund: Rs. 518.40

Passenger 2
Name: Anu
Age: 10
Travel fare: Rs. 480.00
Reservation fee: Rs. 40.00
Ticket total: Rs. 520.00
Cancel this ticket? (yes/no): no

Original booking total: Rs. 1136.00
Total refund: Rs. 518.40
Net amount after refunds: Rs. 617.60
```

7. Retail Billing with Multiple Discount Rules

A 10% discount is allocated proportionally before category-specific taxes.

```
Number of products: 3

Product name: Rice
Category: food
Unit price: 100
Quantity: 5

Product name: Shirt
Category: clothing
Unit price: 1000
Quantity: 2

Product name: Headphones
Category: electronics
Unit price: 3000
Quantity: 1

ITEMIZED INVOICE
Product        Qty    Gross  Discount      Tax     Total
Rice             5   500.00     50.00     0.00    450.00
Shirt            2  2000.00    200.00    90.00   1890.00
Headphones       1  3000.00    300.00   486.00   3186.00

Subtotal: Rs. 5500.00
Discount: Rs. 550.00
Total tax: Rs. 576.00
Grand total: Rs. 5526.00
```

8. Manufacturing Cost and Profitability Analysis

The target is profit as a percentage of production cost.

```
Number of units to produce: 100
Number of raw materials: 1
Material name: Steel
Quantity per finished unit: 2
Cost per material unit: 50
Steel cost: Rs. 10000.00
Extra material wastage percentage: 10
Total labour hours: 20
Labour cost per hour: 200
Total machine hours: 10
Machine cost per hour: 300
Other batch overheads: 2000
Selling price per finished unit: 250
Target profit percentage on cost: 20

PRODUCTION REPORT
Net material cost                  10000.00
Wastage cost                        1000.00
Labour cost                         4000.00
Machine cost                        3000.00
Other overheads                     2000.00
Total production cost              20000.00
Cost per unit                        200.00
Sales revenue                      25000.00
Profit / loss                       5000.00
Target profit                       4000.00
Required selling price/unit          240.00
Actual profit on cost: 25.00 %
Result: profit target achieved.
```

9. Multi-Currency Travel Budget Analysis

Exchange rates are supplied example inputs, not current market rates.

```
Travel budget in INR: 50000
Number of foreign currencies: 2
Currency code: USD
INR value of one currency unit: 83
Currency code: EUR
INR value of one currency unit: 90
Number of expenses: 4

Expense 1
Category: hotel
Amount in original currency: 300
Currency code: USD
Conversion fee percentage: 2
Converted expense: Rs. 24900.00
Conversion fee: Rs. 498.00
Total expense: Rs. 25398.00

Expense 2
Category: food
Amount in original currency: 100
Currency code: EUR
Conversion fee percentage: 2
Converted expense: Rs. 9000.00
Conversion fee: Rs. 180.00
Total expense: Rs. 9180.00

Expense 3
Category: transport
Amount in original currency: 5000
Currency code: INR
Conversion fee percentage: 0
Converted expense: Rs. 5000.00
Conversion fee: Rs. 0.00
Total expense: Rs. 5000.00

Expense 4
Category: food
Amount in original currency: 50
Currency code: USD
Conversion fee percentage: 2
Converted expense: Rs. 4150.00
Conversion fee: Rs. 83.00
Total expense: Rs. 4233.00

CATEGORY TOTALS
hotel : Rs. 25398.00
food : Rs. 13413.00
transport : Rs. 5000.00

Total spending: Rs. 43811.00
Within budget.
Remaining: Rs. 6189.00
Largest expense category/categories:
- hotel
```

10. Campus Event Financial Planning

The break-even count includes the variable costs of additional participants.

```
Venue cost: 10000
Equipment cost: 5000
Transport cost: 3000
Catering cost per participant: 150
Certificate cost per participant: 20
Expected participant count: 100
Registration fee per participant: 250
Confirmed sponsorship amount: 5000

EVENT FINANCIAL REPORT
Fixed costs: Rs. 18000.00
Variable costs: Rs. 17000.00
Total event cost: Rs. 35000.00
Registration revenue: Rs. 25000.00
Confirmed sponsorship: Rs. 5000.00
Expected deficit: Rs. 5000.00
Additional sponsorship required: Rs. 5000.00
Required contribution per participant: Rs. 300.00
Minimum registrations needed to break even: 163
Additional registrations needed: 63
```